

Informal Conjecture on Property (*) for $d = o(n/\log n)$

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

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Summary and provenance

Catalog id	2207.13651__00
Catalog entry	Informal Conjecture on Property (*) for $d = o(n/\log n)$
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	On random irregular subgraphs
Model verdict	proved (confidence: high)
Model's one-line claim	A half-interval decomposition of the vertex labels, followed by read- d Hölder bounds for uniform spacings, proves Property (*) throughout $d = o(n/\log n)$.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2207.13651__00/output.md
Independent review	None
Model's caveats	Literature priority has not been independently checked; graphs are finite, simple, and undirected.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

Property (*) in the conjectured range

1. Statement and the key observation

Let G be a d -regular graph on n vertices. Independently assign $X_v \sim \text{Unif}[0, 1]$, and retain $uv \in E(G)$ exactly when $X_u + X_v \geq 1$. Write

$$m_k = |\{v : \deg_H(v) = k\}|, \quad \mu = \frac{n}{d+1}.$$

We prove the following.

Theorem For every sequence of d -regular graphs with

$$d = o(n/\log n),$$

with probability tending to 1,

$$\max_{0 \leq k \leq d} |m_k - \mu| = o(\mu).$$

The probability estimates are uniform over the choice of G .

More quantitatively, there is an absolute constant C such that, for all sufficiently large n , whenever $d \geq 64 \log n$,

$$\Pr \left(\max_{0 \leq k \leq d} |m_k - \mu| > C \left(\sqrt{\frac{n}{d} \log n} + \log n \right) \right) \leq Cn^{-3}. \quad (1)$$

The useful idea is to record whether each label belongs to the lower or upper half of $[0, 1]$. Conditional on these bits:

- every lower–lower edge is absent;
- every upper–upper edge is present;
- only edges across the resulting bipartition remain random.

Thus one can use the independent-side argument from the supplied earlier attempt **without properly coloring** G . The additional random conditional mean is a sum of bounded local functions of independent bits, and a read- $(d+1)$ Hölder inequality controls it.

We verify all these steps below.

2. Two elementary tools

2.1. Product-space Hölder inequality Suppose $Z_1, \dots, Z_N$ are independent, and each nonnegative function f_i depends on some subset of these coordinates. If every coordinate occurs in at most R functions, then

$$\mathbb{E} \prod_i f_i \leq \prod_i (\mathbb{E} f_i^R)^{1/R}. \quad (2)$$

For completeness, integrate one coordinate at a time. At a coordinate occurring in $q \leq R$ factors, apply Hölder with exponent R to those q factors, adding a constant factor when $q < R$. The resulting factors have the form

$$(\mathbb{E}_{Z_j} f_i^R)^{1/R}.$$

Repeating over the remaining coordinates gives (2).

In particular, if Y_i are functions with this read- R property, then

$$\log \mathbb{E} e^{\lambda \sum_i (Y_i - \mathbb{E} Y_i)} \leq \frac{1}{R} \sum_i \log \mathbb{E} e^{\lambda R (Y_i - \mathbb{E} Y_i)}. \quad (3)$$

2.2. Uniform spacings Let B_r be any specified spacing between r independent uniform points in $[0, 1]$, including the two endpoint spacings. Then

$$B_r \sim \text{Beta}(1, r), \quad \mathbb{E} B_r = \frac{1}{r+1},$$

and, for every integer $j \geq 1$,

$$\mathbb{E} B_r^j = \frac{j!}{(r+1)(r+2) \cdots (r+j)}. \quad (4)$$

These identities follow directly from the constant density of the spacing vector on the simplex.

If $r \geq d/4$, then

$$\mathbb{E}(dB_r)^j \leq 4^j j!.$$

Consequently, for $|\theta| \leq 1/8$, expansion of the exponential gives

$$\begin{aligned} \log \mathbb{E}e^{\theta dB_r} &\leq \mathbb{E}e^{\theta dB_r} - 1 \\ &\leq \theta \frac{d}{r+1} + \sum_{j \geq 2} (4|\theta|)^j \\ &\leq \theta \frac{d}{r+1} + 32\theta^2. \end{aligned} \quad (5)$$

The same bound holds for a variable identically zero, with its mean replacing $d/(r+1)$.

3. Exposing the half-interval bits

Generate the labels as

$$X_v = \frac{C_v + U_v}{2},$$

where all $C_v \sim \text{Bernoulli}(1/2)$ and $U_v \sim \text{Unif}[0, 1]$ are mutually independent.

Put

$$V_a = \{v : C_v = a\}, \quad a \in \{0, 1\}.$$

For each vertex, define its degree across this random cut by

$$r_v = |\{u \in N_G(v) : C_u \neq C_v\}|,$$

and define

$$b_v = \begin{cases} 0, & C_v = 0, \\ d - r_v, & C_v = 1. \end{cases}$$

Almost surely, the degree in H is

$$\deg_H(v) = b_v + \sum_{\substack{u \in N_G(v) \\ C_u \neq C_v}} \mathbf{1}_{\{U_u + U_v \geq 1\}}. \quad (6)$$

Indeed, monochromatic edges are deterministic after exposing the bits, while a cross-edge is retained exactly when $U_u + U_v \geq 1$.

For a fixed bit configuration, let

$$q_{v,k}(C) = \frac{\mathbf{1}_{\{b_v \leq k \leq b_v + r_v\}}}{r_v + 1}, \quad Q_k(C) = \sum_v q_{v,k}(C). \quad (7)$$

The spacing identity, or integration of a binomial distribution, shows that

$$q_{v,k}(C) = \Pr(\deg_H(v) = k \mid C).$$

Thus

$$Q_k(C) = \mathbb{E}[m_k \mid C]. \quad (8)$$

We will use the event

$$\mathcal{E} = \{r_v \geq d/4 \text{ for every } v\}.$$

Conditional on either value of C_v , the variable r_v is $\text{Bin}(d, 1/2)$. The elementary binomial lower-tail bound therefore gives

$$\Pr(r_v < d/4) \leq e^{-d/16}, \quad \Pr(\mathcal{E}^c) \leq ne^{-d/16}. \quad (9)$$

4. Concentration conditional on the bits

Fix a bit configuration belonging to $\mathcal{E}$, and fix k . Define

$$M_{a,k} = \sum_{v \in V_a} \mathbf{1}_{\{\deg_H(v)=k\}}, \quad Q_{a,k} = \sum_{v \in V_a} q_{v,k}.$$

Thus $m_k = M_{0,k} + M_{1,k}$ and $Q_k = Q_{0,k} + Q_{1,k}$.

We first bound the moment-generating function of $M_{a,k}$.

Condition additionally on all U_u with $u \in V_{1-a}$. The degree indicators for vertices in V_a are then independent: by (6), each is a function only of its own remaining label U_v .

Let their conditional success probabilities be $p_{v,k}$. If

$$0 \leq k - b_v \leq r_v,$$

then $p_{v,k}$ is the corresponding spacing among the r_v points

$$(1 - U_u)_{\substack{u \in N_G(v) \\ C_u \neq C_v}}.$$

Otherwise $p_{v,k} = 0$. Consequently,

$$\mathbb{E}[p_{v,k} \mid C] = q_{v,k}, \quad (10)$$

and, in the nonzero case, its marginal conditional distribution is Beta(1, r_v).

For real λ , set $\eta = e^\lambda - 1$. Conditional independence gives

$$\begin{aligned} \mathbb{E}[e^{\lambda M_{a,k}} \mid C] &= \mathbb{E}_{U_{V_{1-a}}} \prod_{v \in V_a} (1 + \eta p_{v,k}) \\ &\leq \mathbb{E}_{U_{V_{1-a}}} \exp \left(\eta \sum_{v \in V_a} p_{v,k} \right). \end{aligned} \quad (11)$$

This inequality is valid for either sign of λ , since $\eta > -1$.

Each opposite-side label is used in at most d of the functions $p_{v,k}$. Applying (2) with $R = d$, and then (5), yields, for $|\eta| \leq 1/8$,

$$\begin{aligned} \log \mathbb{E}[e^{\lambda M_{a,k}} \mid C] &\leq \frac{1}{d} \sum_{v \in V_a} \log \mathbb{E}[e^{\eta d p_{v,k}} \mid C] \\ &\leq \eta Q_{a,k} + 32 \frac{|V_a|}{d} \eta^2. \end{aligned} \quad (12)$$

On $\mathcal{E}$,

$$Q_{a,k} \leq \frac{4|V_a|}{d}.$$

Since $e^\lambda - 1 - \lambda = O(\lambda^2)$, equation (12) implies that, for absolute constants $C_0, \lambda_0 > 0$,

$$\log \mathbb{E}[e^{\lambda(M_{a,k} - Q_{a,k})} \mid C] \leq C_0 \frac{|V_a|}{d} \lambda^2, \quad |\lambda| \leq \lambda_0. \quad (13)$$

The two sides need not be independent. Cauchy–Schwarz nevertheless gives

$$\begin{aligned} \mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] &\leq (\mathbb{E}[e^{2\lambda(M_{0,k} - Q_{0,k})} \mid C])^{1/2} \\ &\quad \times (\mathbb{E}[e^{2\lambda(M_{1,k} - Q_{1,k})} \mid C])^{1/2}. \end{aligned}$$

Hence, after adjusting absolute constants,

$$\log \mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] \leq C_1 \frac{n}{d} \lambda^2, \quad |\lambda| \leq \lambda_1. \quad (14)$$

Chernoff optimization proves the following estimate, uniformly over every bit configuration in $\mathcal{E}$:

$$\Pr(|m_k - Q_k| \geq t \mid C) \leq 2 \exp \left[-c \min \left\{ \frac{dt^2}{n}, t \right\} \right]. \quad (15)$$

This handles both tails and all degree values, including $k = 0, d$.

5. Concentration of the random conditional mean

We next control $Q_k(C)$. It is important here **not** to condition the independent bits on $\mathcal{E}$.

Instead, define local truncations

$$\bar{q}_{v,k} = q_{v,k} \mathbf{1}_{\{r_v \geq d/4\}}, \quad \bar{Q}_k = \sum_v \bar{q}_{v,k}.$$

For every bit configuration,

$$0 \leq \bar{q}_{v,k} \leq \frac{4}{d}. \quad (16)$$

Moreover, $\bar{q}_{v,k}$ depends only on the bits in $N_G[v]$. Each bit belongs to exactly $d+1$ such closed neighborhoods.

Apply (3) with $R = d+1$, followed by the elementary Hoeffding moment-generating-function bound for a variable in an interval of length $4/d$. For every real λ ,

$$\begin{aligned} \log \mathbb{E} e^{\lambda(\bar{Q}_k - \mathbb{E} \bar{Q}_k)} &\leq \frac{1}{d+1} \sum_v \log \mathbb{E} e^{\lambda(d+1)(\bar{q}_{v,k} - \mathbb{E} \bar{q}_{v,k})} \\ &\leq \frac{n(d+1)}{8} \left(\frac{4}{d} \right)^2 \lambda^2 \\ &\leq 4 \frac{n}{d} \lambda^2. \end{aligned}$$

Therefore

$$\Pr(|\bar{Q}_k - \mathbb{E} \bar{Q}_k| \geq t) \leq 2 \exp \left(-\frac{dt^2}{16n} \right). \quad (17)$$

We also need its mean. Conditional on $X_v = x$, the original degree is $\text{Bin}(d, x)$, so

$$\Pr(\deg_H(v) = k) = \int_0^1 \binom{d}{k} x^k (1-x)^{d-k} dx = \frac{1}{d+1}. \quad (18)$$

Thus $\mathbb{E} m_k = \mathbb{E} Q_k = \mu$. Since $0 \leq q_{v,k} \leq 1$, equations (9) and (18) give

$$0 \leq \mu - \mathbb{E} \bar{Q}_k = \sum_v \mathbb{E} [q_{v,k} \mathbf{1}_{\{r_v < d/4\}}] \leq ne^{-d/16}. \quad (19)$$

Finally, on $\mathcal{E}$,

$$Q_k = \bar{Q}_k \quad \text{for every } k. \quad (20)$$

6. Simultaneous concentration for $d \geq 64 \log n$

Put $B = ne^{-d/16}$. Combining (15), (17), (19), and (20), and taking a union bound over the $d+1$ degree values, gives

$$\Pr \left(\max_{0 \leq k \leq d} |m_k - \mu| \geq 2t + B \right) \leq B + 4(d+1) \exp \left[-c \min \left\{ \frac{dt^2}{n}, t \right\} \right]. \quad (21)$$

Here the bad event $\mathcal{E}^c$, which is common to all k , is counted only once. This inequality is valid for every $d \geq 1$, although it is useful in this form when d is at least a sufficiently large multiple of $\log n$.

Suppose now that $d \geq 64 \log n$. Then

$$B \leq n^{-3}.$$

Choose

$$t = A \left(\sqrt{\frac{n}{d} \log n} + \log n \right),$$

where $A \geq 1$ is a sufficiently large absolute constant. We have

$$\frac{dt^2}{n} \geq A^2 \log n, \quad t \geq A \log n.$$

Thus (21) proves (1).

Under the conjectured hypothesis $d = o(n/\log n)$,

$$\frac{\sqrt{(n/d) \log n} + \log n}{n/(d+1)} = O \left(\sqrt{\frac{d \log n}{n}} + \frac{d \log n}{n} \right) = o(1). \quad (22)$$

This proves Property (*) in the large-degree case.

7. The remaining small degrees

For completeness, we handle all $d < 64 \log n$ directly.

For fixed k , let

$$I_{v,k} = \mathbf{1}_{\{\deg_H(v)=k\}}.$$

Each indicator depends on the independent labels in $N_G[v]$, so these functions have the read- R property with

$$R = d + 1.$$

Each indicator is Bernoulli with mean $p = 1/(d+1)$, by (18). Hence (2) gives

$$\mathbb{E} e^{\lambda m_k} \leq (1 - p + p e^{\lambda R})^{n/R}.$$

The usual elementary Chernoff calculation therefore yields, for $0 < \varepsilon \leq 1$,

$$\Pr(|m_k - \mu| \geq \varepsilon \mu) \leq 2 \exp \left(-\frac{\varepsilon^2 n}{3(d+1)^2} \right). \quad (23)$$

Taking a union bound,

$$\Pr \left(\max_k |m_k - \mu| \geq \varepsilon \mu \right) \leq 2(d+1) \exp \left(-\frac{\varepsilon^2 n}{3(d+1)^2} \right). \quad (24)$$

For $d < 64 \log n$, choose

$$\varepsilon_n = 4(d+1) \sqrt{\frac{\log n}{n}}.$$

Uniformly throughout this degree range, $\varepsilon_n = o(1)$, and (24) tends to zero. The case $d = 0$ is deterministic: $m_0 = n$.

Together with Section 6, this covers every sequence $d = o(n/\log n)$, completing the proof. $\square$

8. Scope and relation to the earlier obstruction

The supplied attempt's read- R argument is valid, but a proper coloring of G is unnecessary. Exposing the half-interval bit of every label makes the **random part** of the model bipartite, regardless of $\chi_f(G)$. The monochromatic edges contribute only the deterministic offsets b_v .

The remaining conditional mean is controlled because, outside exponentially unlikely local cut imbalances, each of its summands is at most $4/d$ and reads only $d + 1$ independent bits. This removes the chromatic-number loss and avoids the unresolved residual estimate in the earlier approach.

There is no remaining mathematical gap for the stated simple-graph model. The argument does not establish failure beyond the conjectured range, and literature priority remains unchecked.